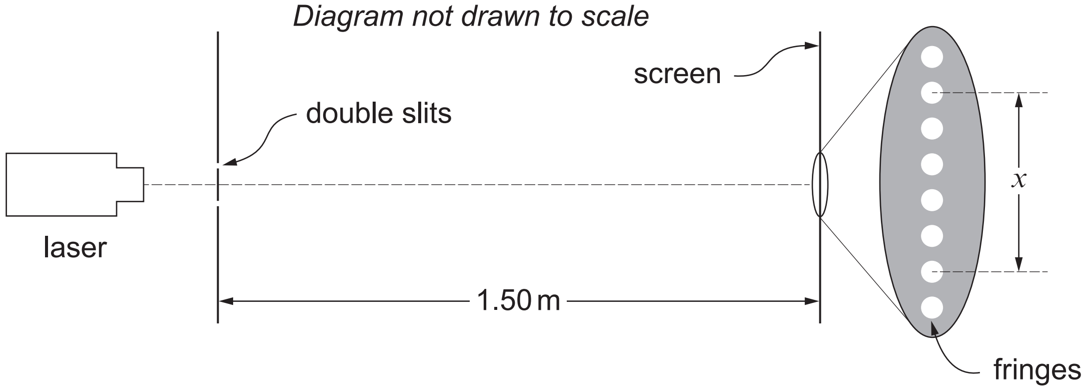
Examiner only

4. *(a)* Explain the conditions that are required of the light sources, for a two-source interference pattern to be observed. Include examples of when the requirements would and would not be met. [6 QER]

Examiner only

(b) A Young's fringes experiment is set up as shown.

Diagram not drawn to scale



(i) Six students measure the distance x, obtaining these results:

6.5 mm 6.3 mm 6.9 mm 6.9 mm 6.7 mm 6.4 mm

Calculate the mean value for fringe separation (the separation of the centres of **neighbouring** bright fringes), together with its **percentage** uncertainty. [4]

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(ii) The uncertainty in the distance of 1.50 m shown in the diagram is negligible. The separation between the centres of the slits is given by the makers of the double slit as 0.60 mm ±5 %. Calculate a value for the wavelength of the light from the laser, along with its **percentage** uncertainty. [3]

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QUESTION CONTINUES ON NEXT PAGE

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(c) The laser beam is now shone normally on to a diffraction grating with centres of slits separated by 1 500 nm. Second order beams emerge at angles of 45° to the normal.

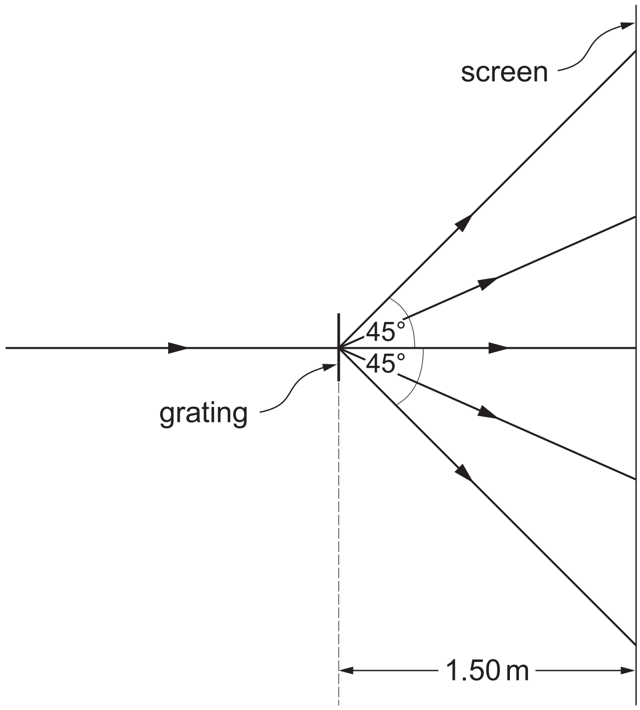

Diagram not drawn to scale

(i) Determine a value for the wavelength of the laser light from the data. [2]

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(ii) The beams from the grating strike the screen as shown opposite. The pattern on the screen may be compared with the pattern of fringes in part *(b)* (Young's experiment). There are fewer bright spots in the case of the grating. State two other differences between the patterns **and** state briefly why each difference occurs. [4]

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